

The complement system

- The complement system is a multi-component system that functions to eliminate microbes.
- The Complement is composed of more than **25 different proteins** produced by different tissues and cells including hepatocytes, macrophages and gut epithelial cells
 - C1(qrs), C2, C3, C4, C5, C6, C7, C8, C9
 - factors B, D, H and I, properdin (P)
 - mannose binding lectin (MBL), MBL associated serine proteases (MASP-1 MASP-2)
 - C1 inhibitor (C1-INH, serpin), C4-binding protein (C4-BP), decay accelerating factor (DAF), Complement receptor 1 (CR1), protein-S (vitronectin)

Functions of Complement

1. Direct Lysis of bacteria, some viruses and cells.
2. Opsonization and enhanced phagocytosis.
3. Triggers cellular functions that contribute to inflammation.
 - Chemokine, histamine release due to complement products called anaphylatoxins
4. Facilitates removal of immune complexes in liver and spleen.

- Activation of complement involves the sequential proteolysis of proteins
- Peptide fragments by activation of a component are denoted by small letters
- Letter “b” is *usually* added to the larger, membrane-binding, peptide and “a” to the smaller peptide (*e.g.*, C3b/C3a, C4b/C4a, C5b/C5a), **EXCEPT** C2 (the larger, membrane-binding moiety is C2a; the smaller one is C2b)

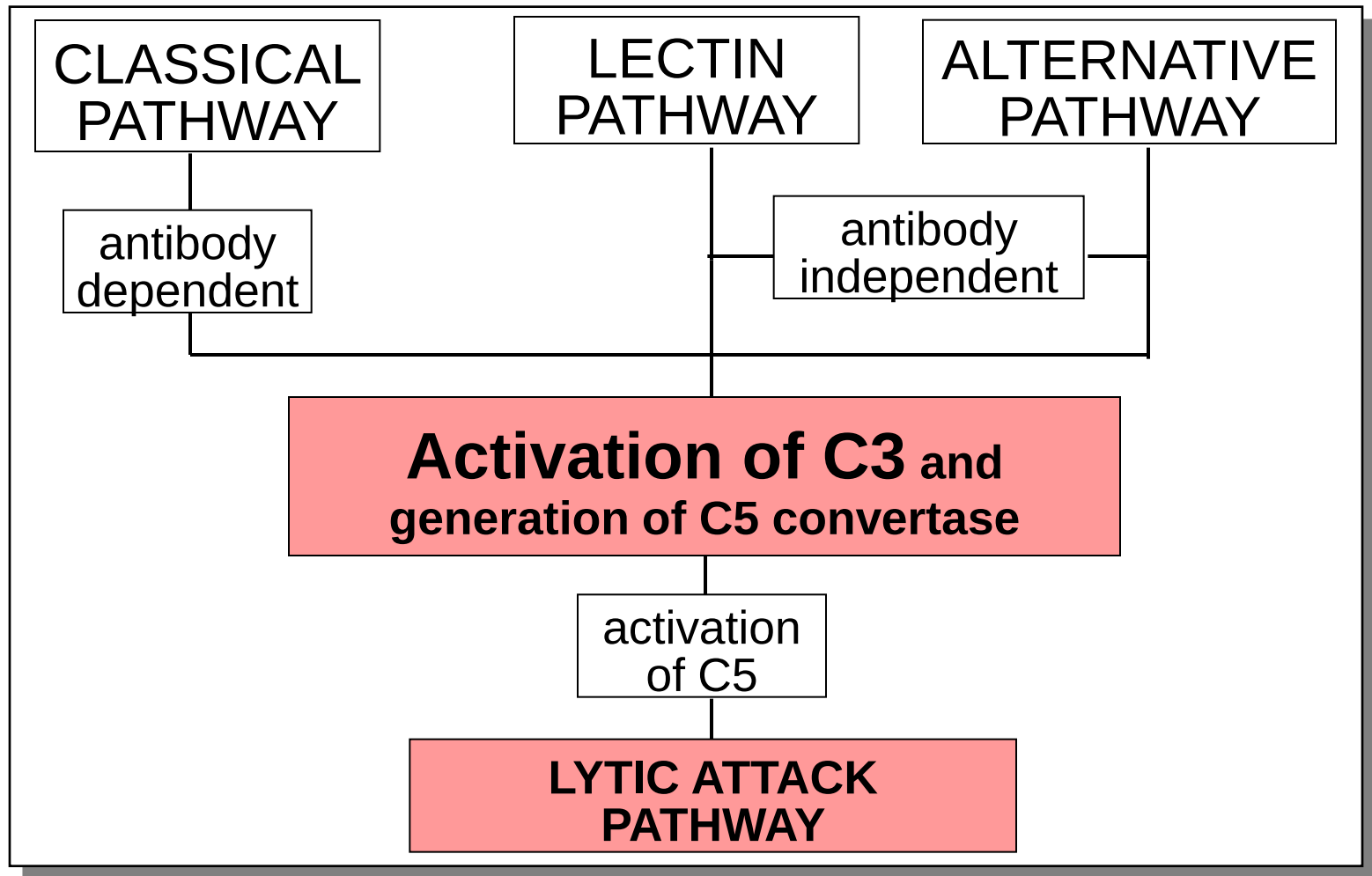
- The smaller fragments usually diffuse away and may bind to cell surface receptors to trigger localized inflammatory response= anaphylatoxins.
- The larger fragment usually remains bound to target cells and facilitate lysis.
- Those complexes that have enzymatic activity are designated by a bar over the number or symbol (e.g., C4b2a, C3bBb)
- Once activated a proteolytic cascade is initiated that ultimately results in complement activation

Complement Activation

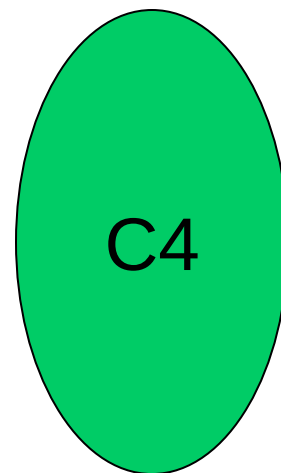
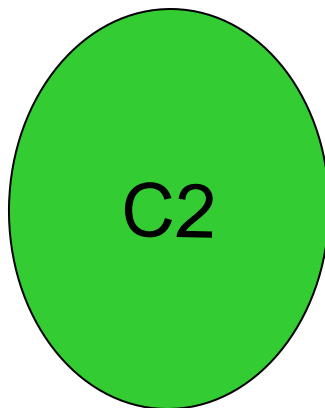
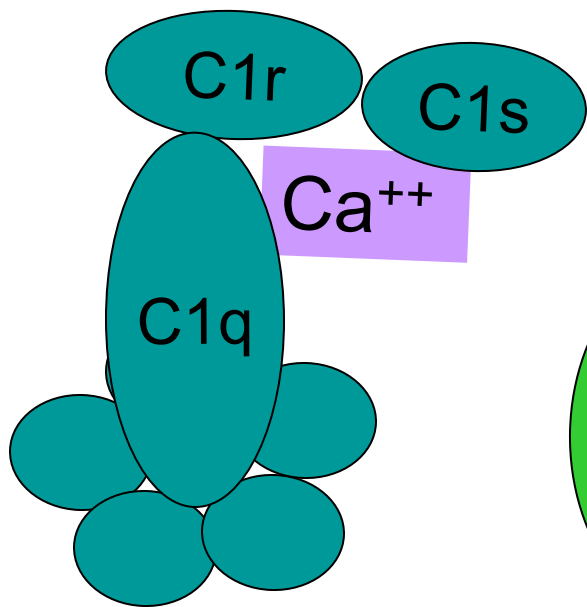
Three major pathways lead to complement activation

1. **Classical pathway**-initiated by antibody binding to antigens on pathogens.
 2. **Alternate pathway** - initiated by direct interaction of complement with pathogen surfaces.
 3. **Lectin pathway** -initiated by binding to mannose residues found on certain bacterial strains.
- All leading to the activation of C5 that follows the activation of the **membrane attack (lytic) pathway**.

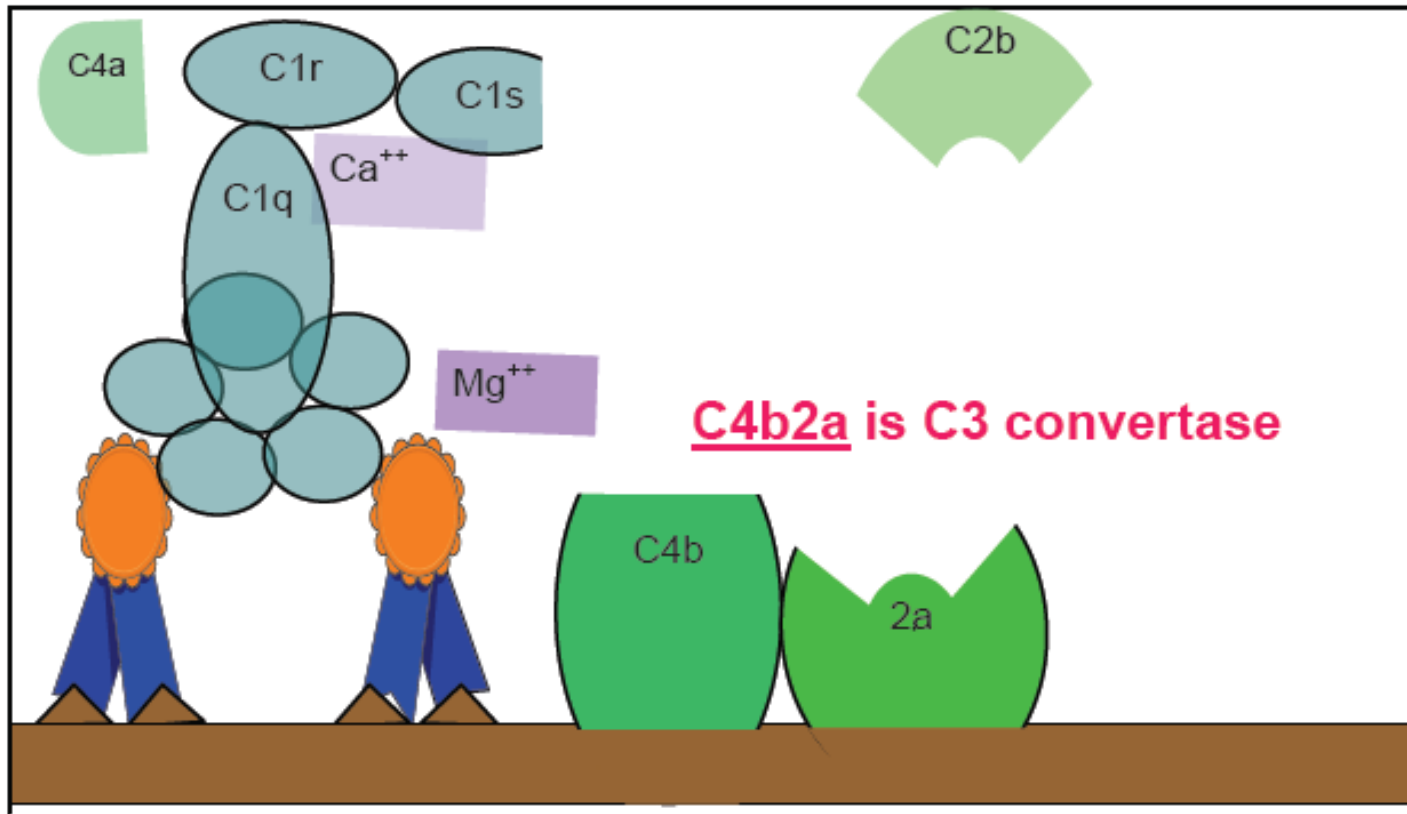
Pathways of complement activation



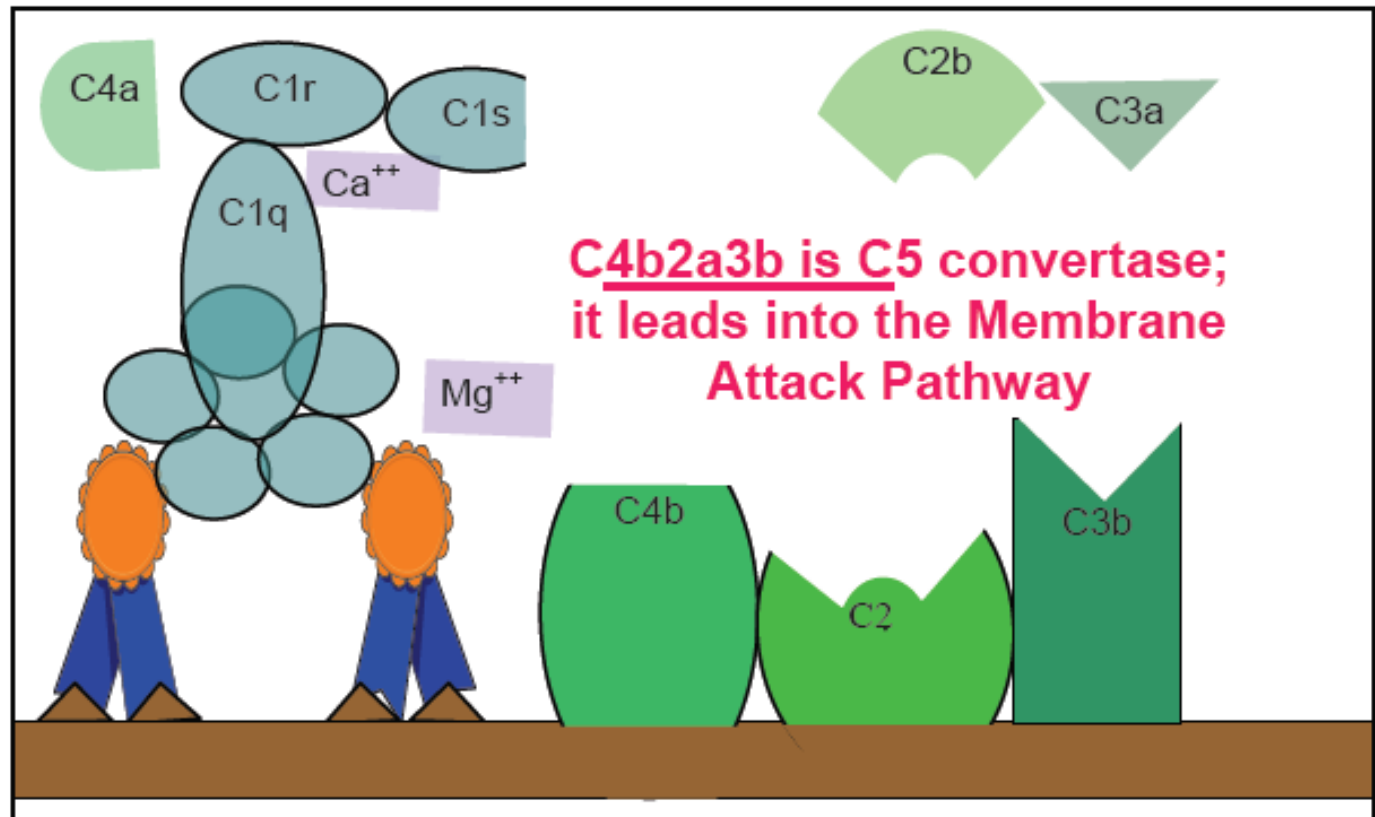
- C1 a multi-subunit protein containing three different proteins C1q, C1r and C1s, binds to the Fc region of IgG and IgM antibody molecules that have interacted with antigen.
- C1 binding does not occur to antibodies that have not complexed with antigen and binding requires calcium and magnesium ions.
- In some cases C1 can bind to aggregated immunoglobulin [e.g. aggregated IgG] or to certain pathogen surfaces in the absence of antibody).



- The binding of C1 to antibody is via C1q and C1q must cross link at least two antibody molecules before it is firmly fixed.
- The binding of C1q results in the activation of C1r which in turn activates C1s.
- The result is the formation of an activated “C1qrs”, which is an enzyme that cleaves C4 into two fragments C4a and C4b.
- The C4b fragment binds to the membrane and the C4a fragment is released into the microenvironment
- Activated “C1qrs” also cleaves C2 into C2a and C2b. C2a binds to the membrane in association with C4b and C2b is released into the microenvironment



- The resulting C4bC2a complex is a C3 convertase, which cleaves C3 into C3a and C3b.
- C3b binds to the membrane in association with C4b and C2a and C3a is released into the microenvironment.
- The resulting C4bC2aC3b is a C5 convertase.
- The generation of C5 convertase is the end of the classical pathway



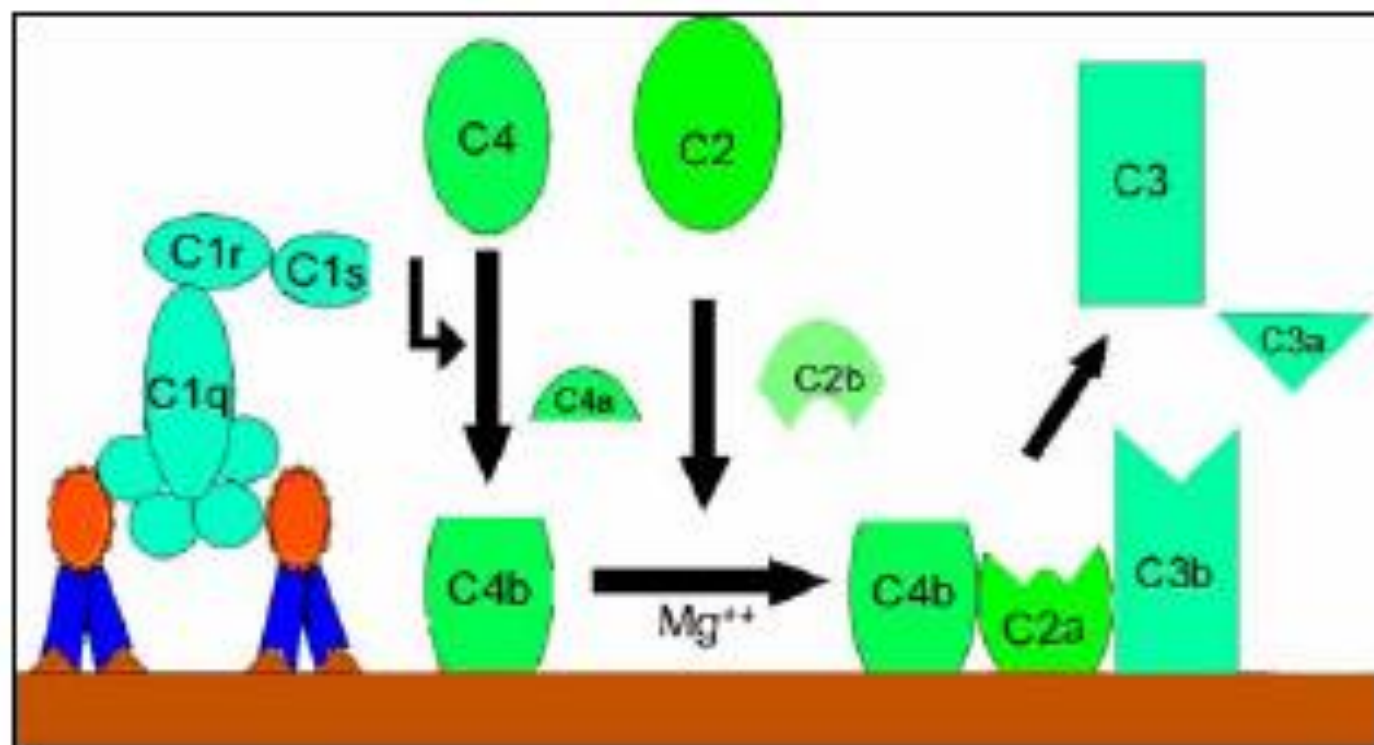


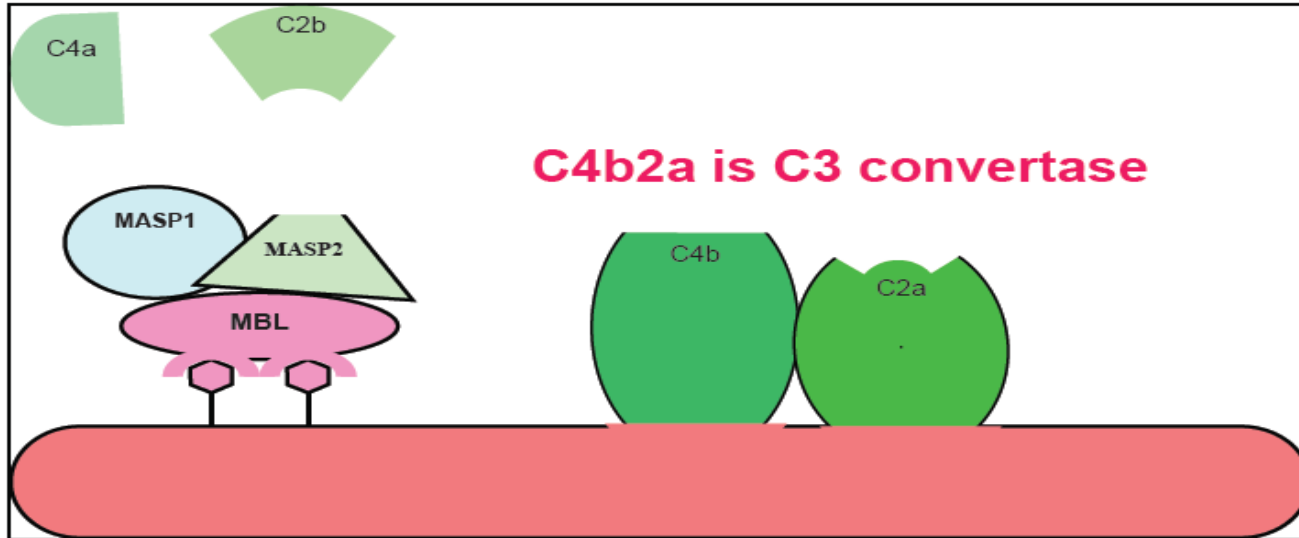
Figure 1. Activation of C3 by the classical pathway

Lectin pathway:

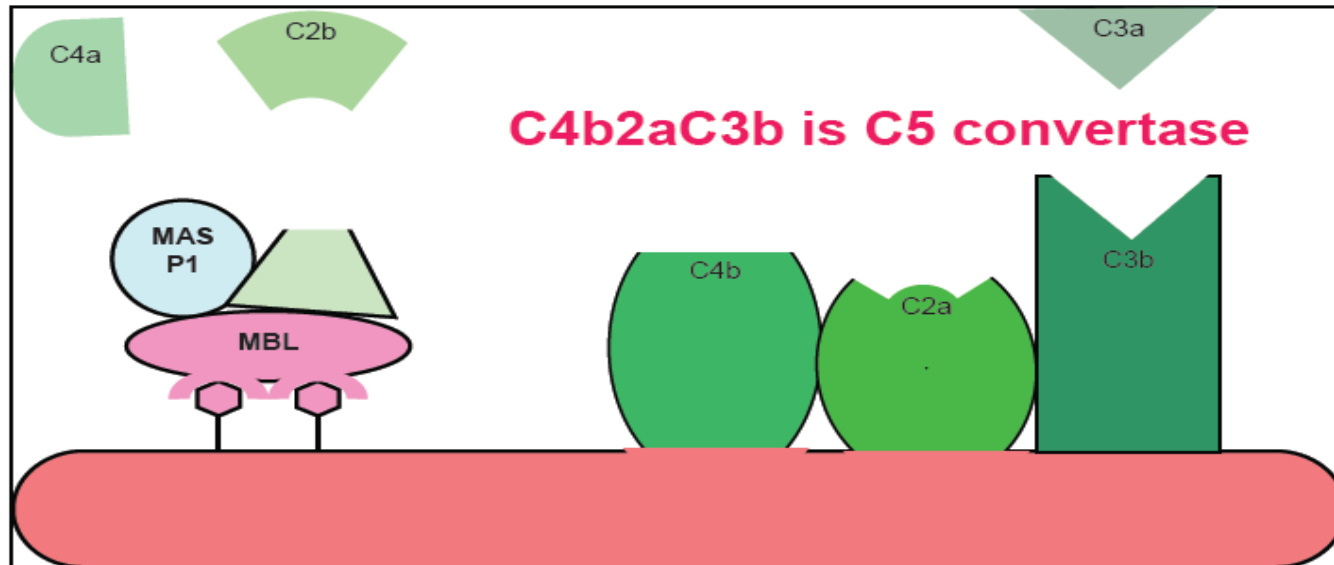
- The lectin pathway is very similar to the classical pathway.
- It is initiated by the binding of mannose binding lectin (MBL) to bacterial surfaces with mannose-containing polysaccharides.
- Binding of MBL to a pathogen results in the association of two serine proteases, MASP-1 and MASP-2 (MBL-associated serine proteases). MASP-1 and MASP-2 are similar to C1r and C1s, respectively and MBL is similar to C1q.
- Formation of the MBL/MASP-1/MASP-2 tri-molecular complex results in the activation of the MASPs and subsequent cleavage of C4 into C4a and C4b.

- The C4b fragment binds to the membrane and the C4a fragment is released into the microenvironment.
- Activated MASPs also cleave C2 into C2a and C2b.
- C2a binds to the membrane in association with C4b and C2b is released into the microenvironment.
- The resulting C4bC2a complex is a C3 convertase, which cleaves C3 into C3a and C3b. C3b binds to the membrane in association with C4b and C2a and C3a is released into the microenvironment.
- The resulting C4bC2aC3b is a C5 convertase. The generation of C5 convertase is the end of the lectin pathway.
- .

Generation of C3 convertase in the lectin pathway



Generation of C5 convertase in the lectin pathway



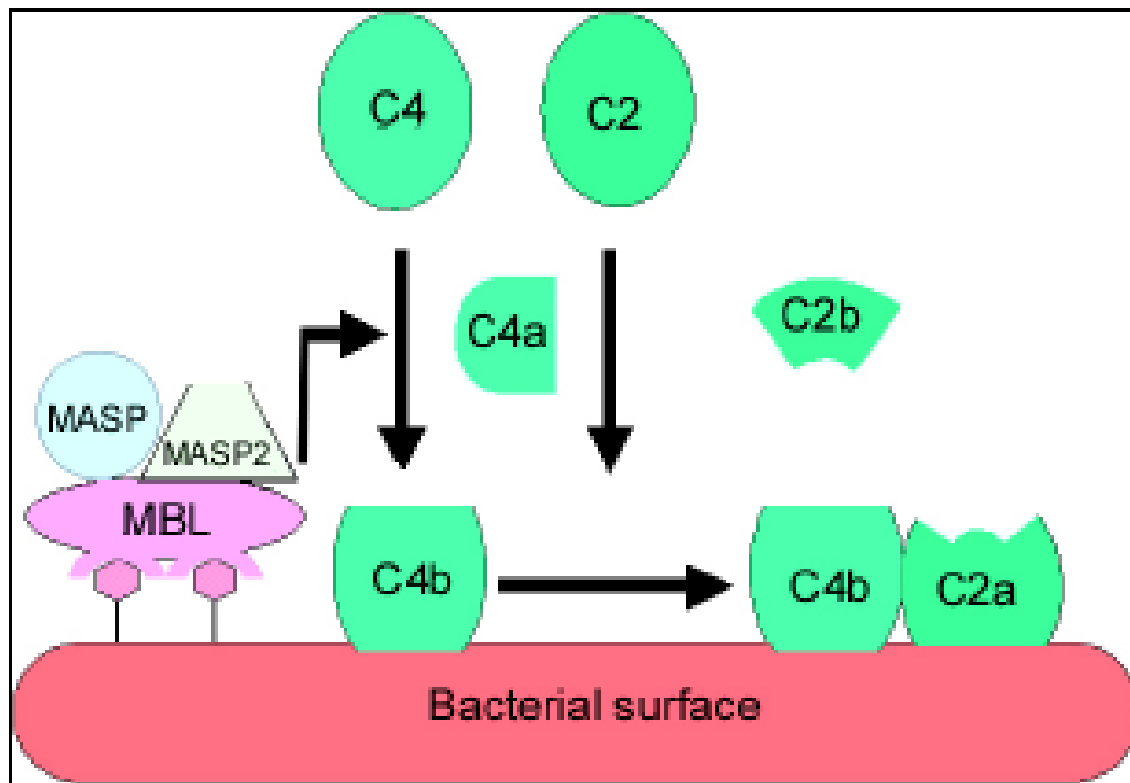


Figure 2. Lectin initiated pathway

Alternative Pathway:

- This major pathway of complement activation involves four serum proteins: C3, factor B, factor D, and properdin
- The alternative pathway is initiated in most cases by cell-surface constituents that are foreign to the host
- In the alternative pathway, serum C3, which contains an unstable thioester bond, is subject to slow spontaneous hydrolysis to yield C3a and C3b.
- The C3b present on the surface of the foreign cells can bind another serum protein called factor B to form a complex

- Once C3b is formed, Factor B will bind to it and becomes susceptible to cleavage by Factor D.
- Factor D cleaves the C3b-bound factor B, releasing a small fragment (Ba) that diffuses away and generating C3bBb.
- The resulting C3bBb complex is a C3 convertase that will continue to generate more C3b, thus amplifying C3b production
- The C3 convertase activity of C3bBb has a half-life of only 5 minutes unless the serum protein **properdin** binds to it, stabilizing it and extending the half-life of this convertase activity to 30 minutes.

- The C3bBb generated in the alternative pathway can activate unhydrolyzed C3 to generate more C3b autocatalytically.
- The C3 convertase activity of C3bBb generates the C3bBb3b complex, which exhibits C5 convertase activity, analogous to the C4b2a3b complex in the classical pathway

Membrane attack pathway

- The terminal sequence of complement activation involves C5b, C6, C7, C8, and C9, which interact sequentially to form a macromolecular structure called the **membrane-attack complex (MAC)**.
- This complex forms a large channel through the membrane of the target cell, enabling ions and small molecules to diffuse freely across the membrane
- C5 convertase from the classical (C4b2a3b), lectin (C4b2a3b) or alternative (C3bBb3b) pathway cleaves C5 into C5a and C5b.

- C5a remains in the fluid phase and the C5b rapidly associates with C6 and C7 and inserts into the membrane. Subsequently C8 binds, followed by several molecules of C9.
- The C9 molecules form a pore in the membrane through which the cellular contents leak and lysis occurs
- Lysis is not an enzymatic process it is thought to be due to physical damage to the membrane. The complex consisting of C5bC6C7C8C9 is referred to as the membrane attack complex (MAC)

Effect	Complement product mediating*
Cell lysis	C5b-9, the membrane-attack complex (MAC)
Inflammatory response	
Degranulation of mast cells and basophils [†]	C3a, C4a, and C5a (anaphylatoxins)
Degranulation of eosinophils	C3a, C5a
Extravasation and chemotaxis of leukocytes at inflammatory site	C3a, C5a, C5b67
Aggregation of platelets	C3a, C5a
Inhibition of monocyte/macrophage migration and induction of their spreading	Bb
Release of neutrophils from bone marrow	C3c
Release of hydrolytic enzymes from neutrophils	C5a
Increased expression of complement receptors type 1 and 3 (CR1 and CR3) on neutrophils	C5a
Opsonization of particulate antigens, increasing their phagocytosis	C3b, C4b, iC3b
Viral neutralization	C3b, C5b-9 (MAC)
Solubilization and clearance of immune complexes	C3b

Receptor	Structure	Ligands	Cell distribution	Function
Type 1 complement receptor (CR1, CD35)	160-250 kD; multiple CCPRs	C3b > C4b > iC3b	Mononuclear phagocytes, neutrophils, B and T cells, erythrocytes, eosinophils, FDCs	Phagocytosis Clearance of immune complexes Promotes dissociation of C3 convertases by acting as cofactor for cleavage of C3b,C4b
Type 2 complement receptor (CR2, CD21)	145-kD; multiple CCPRs	C3d, C3dg > iC3b	B lymphocytes, FDCs, nasopharyngeal epithelium	Coreceptor for B cell activation Trapping of antigens in germinal centers Receptor for EBV
Type 3 complement receptor (CR3, Mac-1, CD11b/ CD18)	Integrin, with 165-kD α chain and 95-kD β 2 chain	iC3b, ICAM-1; also binds microbes	Mononuclear phagocytes, neutrophils, NK cells	Phagocytosis Leukocyte adhesion to endothelium (via ICAM-1)
Type 4 complement receptor (CR4, p150/95, CD11c/CD18)	Integrin, with 150-kD α chain and 95-kD β 2 chain	iC3b	Mononuclear phagocytes, neutrophils, NK cells	Phagocytosis, cell adhesion?

Comparison of Classical, Lectin and Alternative Pathways

	Classical Pathway	Lectin Pathway	Alternative Pathway
Components	C1, C4, C2, & C3	MBL, MASP-1, MASP-2, C4, C2, & C3	C3, Factor B, Factor D & Properdin
Antibody initiated	Yes	No	No
Initiated by pathogen surfaces	Yes (in some cases)	Yes	Yes
Divalent cation required	Yes	Yes	Yes
Prokinin generation	Yes	Yes	No
Anaphylotoxin generation	Yes	Yes	Yes
C3 convertase	C4bC2b	C4bC2b	C3bBb
C5 convertase	C4bC2bC3b	C4bC2bC3b	C3bBbC3b
Feeds into membrane attack pathway	Yes	Yes	Yes
Regulatory Components	C1-INH, C4-BP, Factor I	C1-INH, C4-BP, Factor I	Factor H, Factor I, DAF, CR1

Reading Assignment

✓ Regulation of the Complement System

■